EXPERIMENT 8: ONTOLOGY DEVELOPMENT USING PROTÉGÉ

Aim:

To create a simple university ontology with classes, properties, and individuals.

Software Required:

- Protégé Desktop
- A compatible OWL reasoner, such as HermiT

Procedure / Algorithm:

1. Create a new ontology named UniversityOntology.
2. Create the classes Person, Student, Faculty, and Course.
3. Make Student and Faculty subclasses of Person.
4. Create an object property named enrolledIn with domain Student and range Course.
5. Create individuals Arun (Student) and PythonProgramming (Course).
6. Assert that Arun is enrolled in PythonProgramming.
7. Run the reasoner and inspect the inferred class membership.

Implementation Steps:

1. Open Protégé and create a new ontology named UniversityOntology.
2. Create the classes Person, Student, Faculty, and Course.
3. Set Student and Faculty as subclasses of Person.
4. Create the object property enrolledIn.
5. Set the domain to Student and the range to Course.
6. Create the individual Arun as a Student.
7. Create the individual PythonProgramming as a Course.
8. Add the assertion that Arun enrolledIn PythonProgramming.
9. Run a compatible reasoner and inspect the inferred class membership.

Program:

```
# University Ontology
```

```
# Classes
classes = {
    "Person": [],
    "Student": ["Person"],
    "Faculty": ["Person"],
    "Course": []
}

# Individuals
individuals = {
    "Arun": "Student",
    "PythonProgramming": "Course"
}

# Object property
enrolledIn = {
    "Arun": "PythonProgramming"
}

# Display ontology
print("Ontology: UniversityOntology")

print("\nClasses:")
for c in classes:
    print(c)

print("\nIndividuals:")
```

```
for individual, class_name in individuals.items():
    print(individual, "is a", class_name)


# Reasoning
for individual, class_name in individuals.items():
    if class_name in classes:
        parent_classes = classes[class_name]


        for parent in parent_classes:
            print("\nInferred:", individual, "is a", parent)


# Display relationship
print("\nRelationships:")
for student, course in enrolledIn.items():
    print(student, "enrolledIn", course)
```

Sample Input:

Classes:

Person

Student

Faculty

Course

Individuals:

Arun : Student

PythonProgramming : Course

Object property assertion:

Arun enrolledIn PythonProgramming

Expected Output:

```
==== RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-8.py ===
Ontology: UniversityOntology

Classes:
Person
Student
Faculty
Course

Individuals:
Arun is a Student
PythonProgramming is a Course

Inferred: Arun is a Person

Relationships:
Arun enrolledIn PythonProgramming
```

Result:

Thus, a university ontology was created, and subclass reasoning was used to infer that Arun is a Person.